

Exp. No: 1	Develop the Test Plan for Testing an E-commerce Web/Mobile
Date:	Application(www.amazon.in)

Aim:

The aim of this experiment is to develop a comprehensive test plan for testing the functionality and usability of the e-commerce web/mobile application www.amazon.in.

Software used

- Application Under Test (AUT): Amazon India (www.amazon.in)
- Operating System: Windows 10/11
- Web Browser: Google Chrome
- Documentation Tool: Microsoft Word or Microsoft Excel
- Test Case Management Tool: Microsoft Excel (or Google Sheets)

Objective

- To understand software testing concepts.
- To identify the important features of an e-commerce application.
- To design test cases based on functional and non-functional requirements.
- To ensure the application meets quality standards before release.

Algorithm:

1. Identify the Scope: Determine the scope of testing, including the features and functionalities that need to be tested.
2. Define Test Objectives: Specify the primary objectives of testing, such as functional testing, usability testing, performance testing, security testing, etc.
3. Identify Test Environment: Define the platforms, browsers, devices, and operating systems on which the application will be tested.
4. Determine Test Deliverables: Decide on the documents and artifacts that will be generated during the testing process, such as test cases, test reports, and defect logs.
5. Create Test Strategy: Develop an overall approach for testing, including the testing techniques, entry and exit criteria, and the roles and responsibilities of the testing team.
6. Define Test Scope and Schedule: Specify the timeline for each testing phase and the scope of testing for each phase.

7. Risk Analysis: Identify potential risks and their impact on the testing process, and devise risk mitigation strategies.
8. Resource Planning: Allocate the necessary resources, including the testing team, hardware, and software required for testing.
9. Test Case Design: Prepare detailed test cases based on the requirements and functionalities of the e-commerce application.
10. Test Data Setup: Arrange test data required for executing the test cases effectively.
11. Test Execution: Execute the test cases and record the test results.
12. Defect Reporting: Document any defects encountered during testing and track their resolution.

Test Plan:

The test plan should cover the following sections:

1. Introduction: Briefly describe the purpose of the test plan and provide an overview of the e-commerce application to be tested.
2. Test Objectives: List the primary objectives of testing the application.
3. Test Scope: Specify the features and functionalities to be tested and any limitations on testing.
4. Test Environment: Describe the hardware, software, browsers, and devices to be used for testing.
5. Test Strategy: Explain the overall approach to be followed during testing.
6. Test Schedule: Provide a detailed timeline for each testing phase.
7. Risk Analysis: Identify potential risks and the strategies to mitigate them.
8. Resource Planning: Specify the resources required for testing.
9. Test Case Design: Include a summary of the test cases developed for the application.
10. Test Data Setup: Describe the process of arranging test data for testing.

11. Defect Reporting: Explain the procedure for reporting and tracking defects.

Test Case Table:

Process	No.	Test Case	Steps	Description	Status	Expected Result	Actual Result	Comment
Test Plan	TC001	Scope of Testing	1. Review the test plan document.	Verify the scope of testing.	Done	The test plan includes all features.		
	TC002	Test Objectives	1. Review the test plan document.	Verify the test objectives.	Done	The test objective sare well-defined.		
	TC003	Test Environment	1. Review the test plan document.	Check the specified environments.	Done	Test environment sare mentioned.		
	TC004	Test Deliverables	1. Review the test plan document.	Ensure all deliverable sare listed.	Done	The test plan includes all deliverables.		
	TC005	Test Strategy	1. Review the test plan document.	Verify the overall approach.	Done	The test strategy is clearly stated.		
	TC006	Test Scope and Schedule	1. Review the test plan document.	Check the schedule and scope.	Done	The schedule and scope are defined.		

	TC007	Risk Analysis	1. Review the test plan document.	Ensure potential risks are identified.	Done	Risks and mitigation strategies are mentioned		
	TC008	Resource Planning	1. Review the test plan document.	Check the required resources	Done	Resources needed for testing are listed.		
	TC009	Test Case Design	1. Review and execute the test cases.	Validate the prepared test cases.	Done	Test cases are accurate and functional.		
	TC010	Test Data Setup	1. Review the test data setup process.	Verify the availability of test data.	Done	Test data is available for testing.		
	TC011	Test Execution	1. Run the test cases and document the outcomes.	Execute the test cases.	In Progress	Test results are recorded and documented		
	TC012	Defect Reporting	1. Log defects with detailed information	Ensure defects are reported correctly.	Not Started	Defects are reported with sufficient details.		
	TC013	Defect Tracking	1. Monitor defect status and updates.	Verify the tracking of defects.	Not Started	Defects are tracked until resolution.		

Explanation:

The test plan is a crucial document that outlines the entire testing process. It ensures that all aspects of the e-commerce application are thoroughly tested, and the results are systematically documented.

Result:

Upon completion of the experiment, you will have a well-structured test plan that provides a clear road map for testing the e-commerce web/mobile application www.amazon.in.

Exp. No: 2

Date:

Design the Test Cases for Testing the E-commerce Application

Aim:

The aim of this experiment is to design a set of comprehensive and effective test cases for testing the e-commerce application www.amazon.in.

Software Used:

- Application Under Test (AUT): Amazon India (www.amazon.in)
- Operating System: Windows 10/11
- Web Browser: Google Chrome
- Documentation Tool: Microsoft Excel / Microsoft Word

Objective:

- To design effective test cases for the Amazon India e-commerce web/mobile application.
- To verify the functionality of key modules such as registration, login, product search, shopping cart, checkout, and payment.
- To ensure the application meets the specified functional requirements by validating expected and actual results.

Algorithm:

1. Understand Requirements: Familiarize yourself with the functional and non-functional requirements of the e-commerce application.
2. Identify Test Scenarios: Based on the requirements, identify different test scenarios that cover all aspects of the application.
3. Write Test Cases: Develop test cases for each identified scenario, including preconditions, steps to be executed, and expected outcomes.
4. Cover Edge Cases: Ensure that the test cases cover edge cases and boundary conditions to verify the robustness of the application.
5. Prioritize Test Cases: Prioritize the test cases based on their criticality and relevance to the application.
6. Review Test Cases: Conduct a peer review of the test cases to ensure their accuracy and completeness.
7. Optimize Test Cases: Optimize the test cases for reusability and maintainability.

Test Case Design:

The test case design should include the following components for each test case:

1. Test Case ID: A unique identifier for each test case.
2. Test Scenario: Description of the scenario being tested.
3. Test Case Description: Detailed steps to execute the test.
4. Precondition: The necessary conditions that must be satisfied before executing the test case.
5. Test Steps: The sequence of actions to be performed during the test.
6. Expected Result: The outcome that is expected from the test.

Test Case Table:

Process	No.	Test Case	Steps	Description	Status	Expected Result	Actual Result	Comment
Test Case Design	TC001	User Registration	1.Navigate to the registration page.	Verify user registration process.	Done	User can successfully register.		
	TC002	User Login	1.Navigate to the login page.	Verify user login process.	Done	User can successfully log in.		
	TC003	Search Functionality	1. Enter a keyword in the search bar.	Verify search functionality	Done	Search results relevant to the keyword.		
	TC004	Add to Cart	1. Browse the product catalog.	Verify adding products to the cart.	Done	Product is added to the shopping cart.		

	TC005	Shopping Cart Validation	1. Click on the shopping cart icon.	Verify the shopping cart contents.	Done	Items in the shopping cart are displayed.		
	TC006	Check out Process	1. Click on the "Checkout" button.	Verify the checkout process.	Not Started	Checkout process proceeds as expected.		

Explanation:

Test cases are designed to validate the functionality and behaviour of the e-commerce application. They ensure that the application performs as intended and meets the specified requirements.

Result:

Upon completion of the experiment, you will have a set of well-defined test cases ready for testing the e-commerce application www.amazon.in.